Google Cloud Storage - Bucket Lock

Step 1: Create the bucket

Open your Google Cloud Console using the temporary lab account, then activate Cloud Shell by clicking the terminal icon >_ at the top.

Run these commands one by one.

1. Set the region

```
gcloud config set compute/region us-west4
```

2. Set the bucket name to your lab Project ID

```
export BUCKET=$(gcloud config get-value project)
```

3. Create the bucket

```
gcloud storage buckets create "gs://$BUCKET"
```

Expected output: a message confirming the bucket was created.

Go back to the lab page and click Check my progress → Create a storage bucket.

Step 2: Set the retention policy

Run:

```
gcloud storage buckets update "gs://$BUCKET" --retention-period=10s
```

Verify the policy:

```
gcloud storage buckets describe "gs://$BUCKET" --
format="default(retention_policy)"
```

Upload the transaction record:

```
gcloud storage cp gs://spls/gsp297/dummy_transactions "gs://$BUCKET/"
```

Check its retention expiration:

```
gcloud storage ls -L "gs://$BUCKET/dummy_transactions"
```

Click Check my progress → Set up Retention Policy.

Step 3: Lock the retention policy

Run:

```
gcloud storage buckets update "gs://$BUCKET" --lock-retention-period
```

When prompted:

```
This setting cannot be reverted! Continue? [y|N]:
```

Type `y` and press Enter.

Important: Locking is permanent. The retention period cannot subsequently be reduced or removed while the bucket exists.

Click Check my progress → Lock the Retention Policy.

Step 4: Configure a temporary hold

Enable the hold:

```
gcloud storage objects update "gs://$BUCKET/dummy_transactions" --
temporary-hold
```

Try deleting the object:

```
gcloud storage rm "gs://$BUCKET/dummy_transactions"
```

The deletion should fail because the object is under a temporary hold.

Release the hold:

```
gcloud storage objects update "gs://$BUCKET/dummy_transactions" --no-
temporary-hold
```

Wait until the 10-second retention period has expired, then delete the object:

```
gcloud storage rm "gs://$BUCKET/dummy_transactions"
```

Click Check my progress → Set up Temporary Hold.

Step 5: Configure event-based holds

Enable the default event-based hold for new objects:

```
gcloud storage buckets update "gs://$BUCKET" --default-event-based-hold
```

Upload a loan record:

```
gcloud storage cp gs://spls/gsp297/dummy_loan "gs://$BUCKET/"
```

Verify the hold:

```
gcloud storage ls -L "gs://$BUCKET/dummy_loan"
```

You should see:

```
Event-Based Hold: TRUE
```

Release the hold:

```
gcloud storage objects update "gs://$BUCKET/dummy_loan" --no-event-based-hold
```

Verify the retention expiration:

```
gcloud storage ls -L "gs://$BUCKET/dummy_loan"
```

Attempt deletion:

```
gcloud storage rm "gs://$BUCKET/dummy_loan"
```

If deletion fails because the retention period has not expired, wait a few seconds and retry.

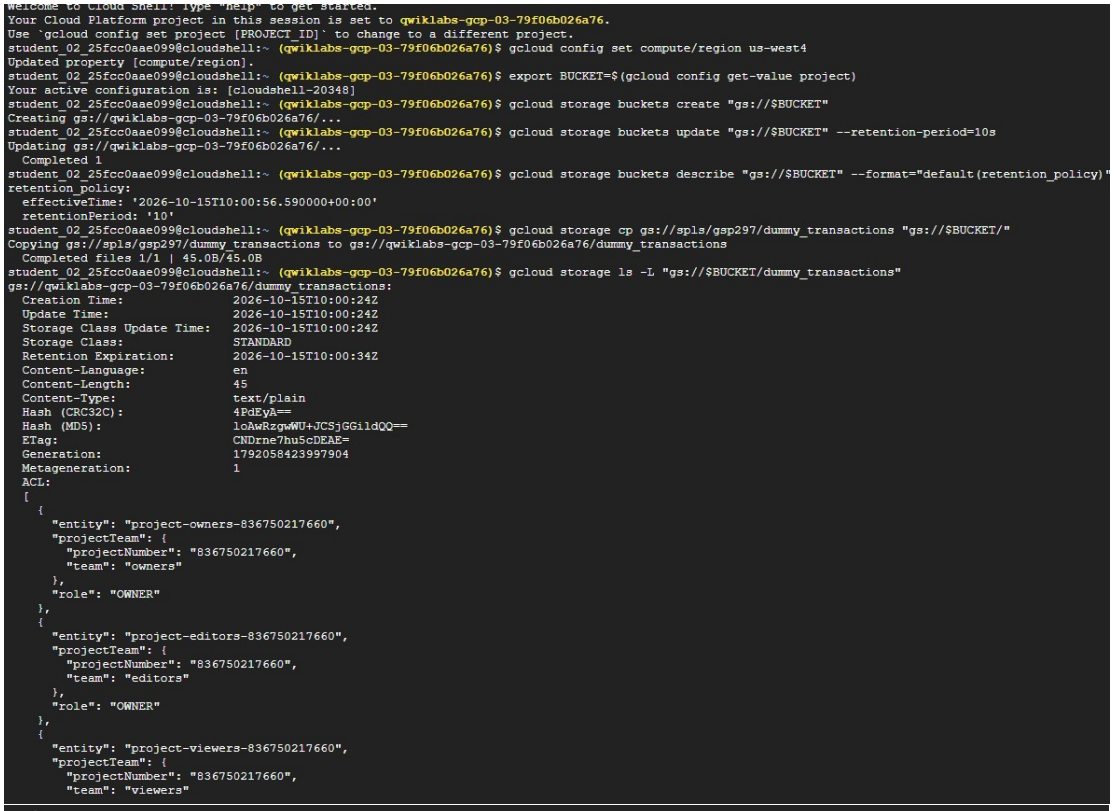
Click Check my progress → Create Event-based holds.

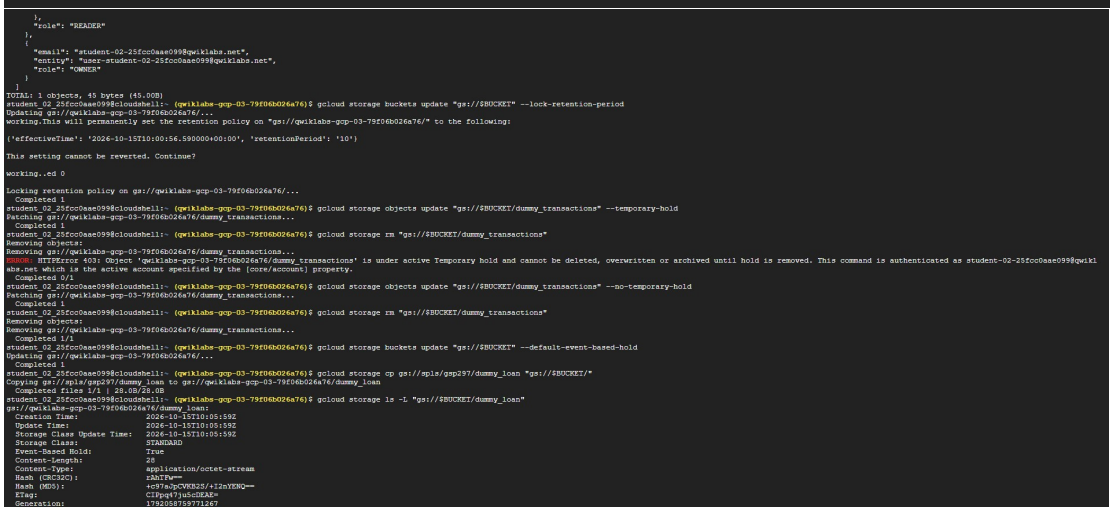
Step 6: Delete the bucket

Once the bucket is empty and no retained objects remain, run:

```
gcloud storage buckets delete "gs://$BUCKET/"
```

This completes the lab's final task.





```
  Generation:                 1792058759771267
  Metageneration:             1
  ACL:
  [
    {
      "entity": "project-owners-836750217660",
      "projectTeam": {
        "projectNumber": "836750217660",
        "team": "owners"
      },
      "role": "OWNER"
    },
    {
      "entity": "project-editors-836750217660",
      "projectTeam": {
        "projectNumber": "836750217660",
        "team": "editors"
      },
      "role": "OWNER"
    },
    {
      "entity": "project-viewers-836750217660",
      "projectTeam": {
        "projectNumber": "836750217660",
        "team": "viewers"
      },
      "role": "READER"
    },
    {
      "email": "student-02-25fcc0aae099@qwiklabs.net",
      "entity": "user-student-02-25fcc0aae099@qwiklabs.net",
      "role": "OWNER"
    }
  ]
TOTAL: 1 objects, 28 bytes (28.00B)
student_02_25fcc0aae099@cloudshell:~ (qwiklabs-gcp-03-79f06b026a76)$ gcloud storage objects update "gs://$BUCKET/dummy_loan" --no-event-based-hold
Patching gs://qwiklabs-gcp-03-79f06b026a76/dummy_loan...
  Completed 1
student_02_25fcc0aae099@cloudshell:~ (qwiklabs-gcp-03-79f06b026a76)$ gcloud storage ls -L "gs://$BUCKET/dummy_loan"
gs://qwiklabs-gcp-03-79f06b026a76/dummy_loan:
  Creation Time:              2026-10-15T10:05:59Z
  Update Time:                2026-10-15T10:06:52Z
  Storage Class Update Time:  2026-10-15T10:05:59Z
  Storage Class:              STANDARD
  Retention Expiration:       2026-10-15T10:07:02Z
  Content-Length:             28
  Content-Type:               application/octet-stream
  Hash (CRC32C):              rAhTFw==
  Hash (MD5):                 +c97aJpCVKB2S/+I2nYENQ==
  ETag:                       CIPpq47ju5cDEAI=
  Generation:                 1792058759771267
  Metageneration:             2
  ACL:
  [
    {
      "entity": "project-owners-836750217660",
      "projectTeam": {
```

```
  ACL:
  [
    {
      "entity": "project-owners-836750217660",
      "projectTeam": {
        "projectNumber": "836750217660",
        "team": "owners"
      },
      "role": "OWNER"
    },
    {
      "entity": "project-editors-836750217660",
      "projectTeam": {
        "projectNumber": "836750217660",
        "team": "editors"
      },
      "role": "OWNER"
    },
    {
      "entity": "project-viewers-836750217660",
      "projectTeam": {
        "projectNumber": "836750217660",
        "team": "viewers"
      },
      "role": "READER"
    },
    {
      "email": "student-02-25fcc0aae099@qwiklabs.net",
      "entity": "user-student-02-25fcc0aae099@qwiklabs.net",
      "role": "OWNER"
    }
  ]
TOTAL: 1 objects, 28 bytes (28.00B)
student_02_25fcc0aae099@cloudshell:~ (qwiklabs-gcp-03-79f06b026a76)$ gcloud storage rm "gs://$BUCKET/dummy_loan"
Removing objects:
Removing gs://qwiklabs-gcp-03-79f06b026a76/dummy_loan...
  Completed 1/1
student_02_25fcc0aae099@cloudshell:~ (qwiklabs-gcp-03-79f06b026a76)$ gcloud storage buckets delete "gs://$BUCKET/"
Removing gs://qwiklabs-gcp-03-79f06b026a76/...
  Completed 1/1
student_02_25fcc0aae099@cloudshell:~ (qwiklabs-gcp-03-79f06b026a76)$
```